

We saw ① Let V, W be fin. dim v. spaces

$T: V \rightarrow W$ be a linear map (i.e. T preserves linear combinations)

If $\dim V$ is finite, $\dim V = \dim \ker T + \dim \operatorname{im} T$

We saw an abstract proof.

② Concrete description of T using bases. Let

$\dim V = p$, ordered basis $(v_1, v_2, \dots, v_p) = B$

$\dim W = q$, ordered basis $(w_1, w_2, \dots, w_q) = C$

We have a recipe to produce a matrix from lin map T :

$$\begin{array}{ccc} \operatorname{Hom}(V, W) & \longrightarrow & q \times p \text{ matrices} \\ T & \longmapsto & \text{matrix } M_{B,C}(T) \end{array}$$

columns of $M_{B,C}(T)$ = coordinate vectors of $T(v_1), T(v_2), \dots, T(v_p)$ w.r.t. C

Exercise (HW) The map $M_{B,C}$ is an isomorphism

• Composition of linear maps in terms of their matrices

$$\left. \begin{array}{ccc} V & \xrightarrow{T} & W \xrightarrow{S} U \\ \dim & p & q \quad r \\ \text{Basis} & B & C \quad D \\ (v_1, \dots, v_p) & (w_1, \dots, w_q) & (u_1, \dots, u_r) \end{array} \right\} \begin{array}{l} \text{Matrix of } S \circ T \text{ in terms} \\ \text{of matrices of } S \text{ \& } T = ? \\ \text{Answer:} \\ M_{B,D}(S \circ T) = M_{C,D}(S) M_{B,C}(T) \end{array}$$

Pf. Let's see 2,3 entry of this matrix
= coeff of u_2 in $S \circ T(v_3)$

- HW 3 may be modified slightly tomorrow
- Quiz next Tue in class
- Attend tutorials

$$S \circ T(v_3) = S \left(\sum_{j=1}^9 t_{j3} w_j \right)$$

Matrix of T : (t_{ij})
 Matrix of S : (s_{ij})

$$= \sum_{j=1}^9 t_{j3} S(w_j)$$

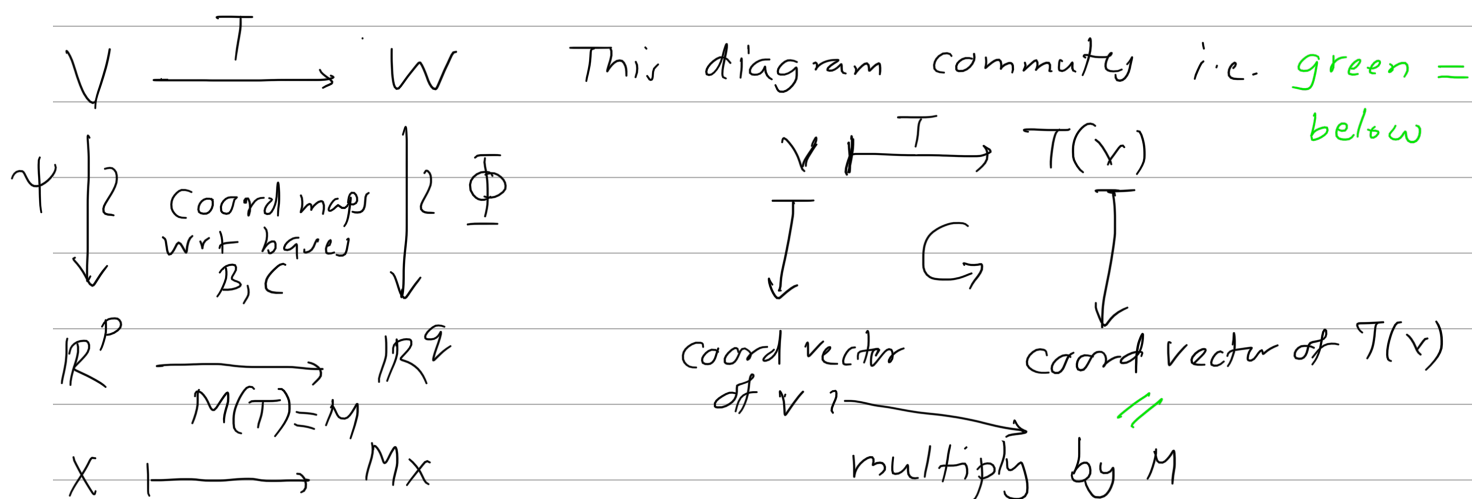
$$= \sum_{j=1}^9 t_{j3} \sum_{i=1}^7 s_{ij} u_i \quad \text{Fix } i=2$$

$$\text{Coeff of } u_2 = \sum_{j=1}^9 s_{2j} t_{j3} = 23 \text{ entry of matrix product } M(S)M(T)$$

You should write a formal proof

according to our defⁿ of matrix mult.

- Concrete proof of dimension formula



$$\dim \ker T + \dim \text{im } T \stackrel{?}{=} \dim V$$

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$$\dim \ker (x \mapsto Mx) + \dim \text{im } (x \mapsto Mx) = p$$

As lower equality was proved (using row reduction) earlier, upper equality follows. ✓

Reason for vertical equalities: show that

$$\Psi(\ker T) = \ker (x \mapsto Mx) \quad \& \quad \Psi \text{ is an isom}$$

$$\Phi(\operatorname{im} T) = \operatorname{im} (x \mapsto Mx) \quad \& \quad \Phi \text{ is an isom}$$

Rmk: We do not yet have an abstract proof of

row rk = col rk. We will see one later using duals.
 We did this using row reduction. Compare with Artin's more indirect proof using dimension formula and the matrix below

- We will study how to look for a good basis for a lin map

Another look at dim formula pf. $V \xrightarrow{T} W$
 $\dim p, \quad q, \quad \operatorname{rk} T = r$

(i) Take a basis of $\ker T$ $k_1, k_2, \dots, k_{p-r}$

(ii) Extend to get a basis of V : $k_1, \dots, k_{p-r}, v_1, \dots, v_r$

(iii) We saw $T(v_1), \dots, T(v_r)$ is a basis for $\operatorname{im} T$

(iv) Extend to get basis $T(v_1), \dots, T(v_r), w_1, w_2, \dots, w_{q-r}$ of W

Matrix of T wrt these bases

$$\left[\begin{array}{c|ccc} & k_1 & k_2 & \dots & k_{p-r} & v_1 & \dots & v_r \\ \hline & & & & & 1 & & 0 \\ & & & & & & \ddots & \\ & & & & & 0 & & 1 \\ & & & & & & & \\ \hline & & & & & 0 & & \end{array} \right] \begin{array}{l} T(v_1) \\ T(v_2) \\ \vdots \\ T(v_r) \\ w_1 \\ \vdots \\ w_{q-r} \end{array}$$

$$T(v_1) = 1 T(v_1) + 0 \cdot (\text{rest}) \text{ etc.}$$

Reordering basis of V I would get matrix of T to be =

$$r = \operatorname{rank} \text{ of } T := \dim \operatorname{im} T$$

$$(*) \left[\begin{array}{c|ccc} & v_1 & \dots & v_r & k_1 & \dots \\ \hline & & & I_{r \times r} & 0 \\ \hline & & & 0 & 0 \end{array} \right]$$

Rmk If T itself is $\mathbb{R}^p \rightarrow \mathbb{R}^q$ then the
 $x \mapsto Ax$

fact that matrix of T wrt some basis looks like $(*)$

can be interpreted as follows: There are
invertible matrices Q & P s.t. $Q^{-1}AP = (*)$

Interesting situation: when $V=W$, a lin map $V \xrightarrow{T} V$
is called an operator on V . Then we'd like to
choose a single basis for V wrt which the matrix
of T becomes "nice". We'll spend a lot of time on this!

We need to set up mechanism for change of basis

Step 1 for single vector space

Suppose V has basis $B = (v_1, \dots, v_p)$
and another basis $B' = (w_1, \dots, w_p)$

Let $u \in V$

$$\left. \begin{aligned} u &= a_1 v_1 + \dots + a_p v_p \\ &= b_1 w_1 + \dots + b_p w_p \end{aligned} \right\} \text{How to go from } \begin{bmatrix} a_1 \\ \vdots \\ a_p \end{bmatrix} \text{ to } \begin{bmatrix} b_1 \\ \vdots \\ b_p \end{bmatrix}$$

We will see this in the next class

Step 2 is to study change of bases for a linear map.
(plural)